Cloud sedimentation × precip suppression: full matrix, 40 km to 1.25 km

6 Nov 2026. Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz. DP-EAMxx DYCOMS-II RF02, 6 h; all 144 runs, single realizations; hour 4–6 means unless noted. Sub-km runs are on hold.

Findings

1. **At ≥5 km, the positive LWP–Nd response is all precip suppression.** With suppression off, LWP is flat at 170–175 $\mathrm{g\,m^{-2}}$ ($S \approx 0$) wherever $w_e \approx 4.9\,\mathrm{mm\,s^{-1}}$.
2. **At coarse resolution, sedimentation switches entrainment between two levels.** At low Nd with sed×1 or ×2, w_e drops to 2.5–2.9 $\mathrm{mm\,s^{-1}}$, with suppression on or off. With suppression off, this gives a negative low-Nd branch (S_{low} from −0.3 to −0.5). Sed×0.5 removes it.
3. **At 2.5 km and 1.25 km the picture is consistent.**
   - Suppression on: S_{low} grows with sed (0.45/0.57/0.71 at 2.5 km; 0.53/0.57/0.70 at 1.25 km).
   - Suppression on: S_{high} is near zero (0.05 at 2.5 km; between −0.03 and +0.02 at 1.25 km).
   - Suppression off: $S \approx 0$, except sed×2 at low Nd, where $S_{\mathrm{low}} \approx 0.25$ because the Nd-20 deck breaks up (cloud cover 0.63–0.69).
   - w_e rises with Nd and falls with the sed factor, with suppression on or off. At 1.25 km and Nd 20 with suppression on, $w_e = 4.1/3.5/2.8\,\mathrm{mm\,s^{-1}}$ for sed ×0.5/×1/×2.
4. **LWP spatial SD** is 1–3 $\mathrm{g\,m^{-2}}$ at ≥10 km, 6–60 at 5 km, and 60–120 at 2.5 and 1.25 km. It is largest at low Nd with sed×2.
5. **The runs have not reached steady state.** LWP is still drifting at hour 6 in many runs (Fig. 3; e.g. 40 km, Nd 20, suppression off). Treat the hour 4–6 means as transient. The 12 h RF02 forcing would test this.

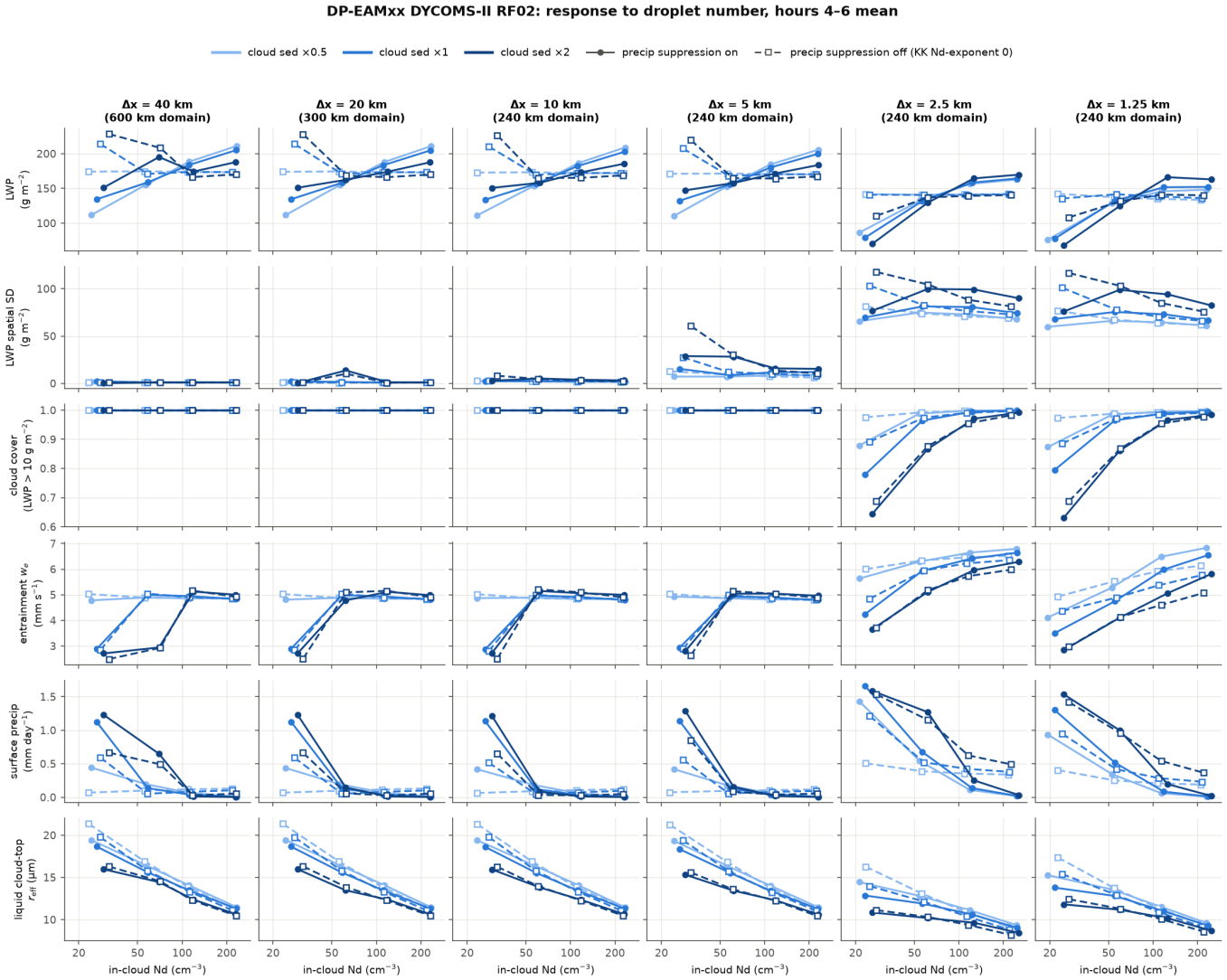


Fig. 1. Response to in-cloud Nd. Columns: resolution. Shade: sed factor. Solid lines with filled markers: suppression on. Dashed lines with open markers: suppression off.

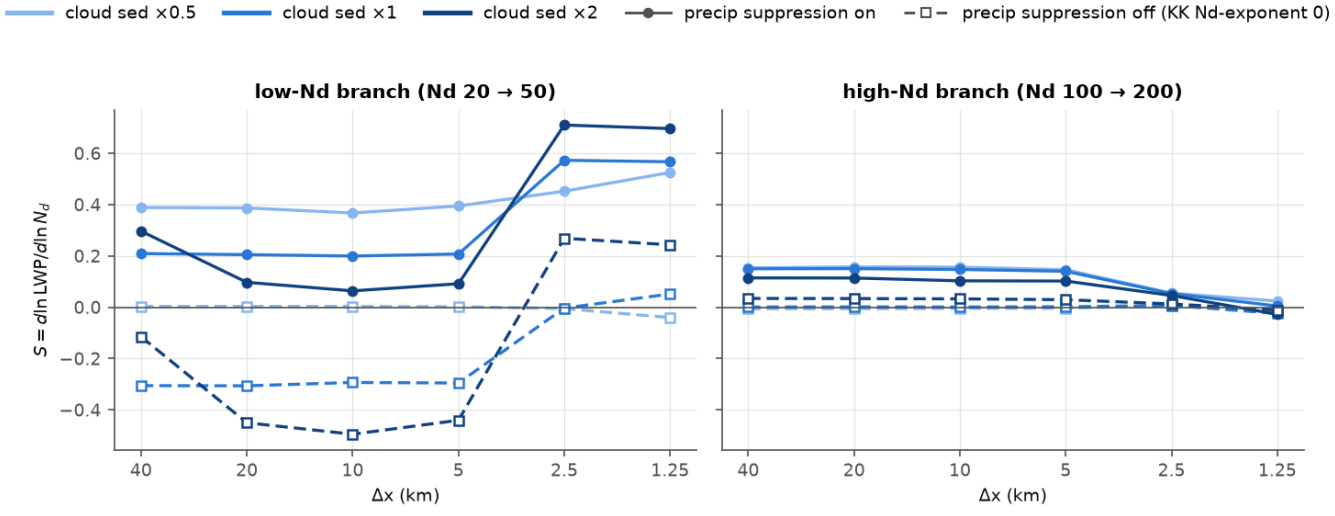


Fig. 2. $S = d\ln \mathrm{LWP}/d\ln N_d$ (diagnosed Nd) versus resolution, for the low-Nd (20→50) and high-Nd (100→200) branches.

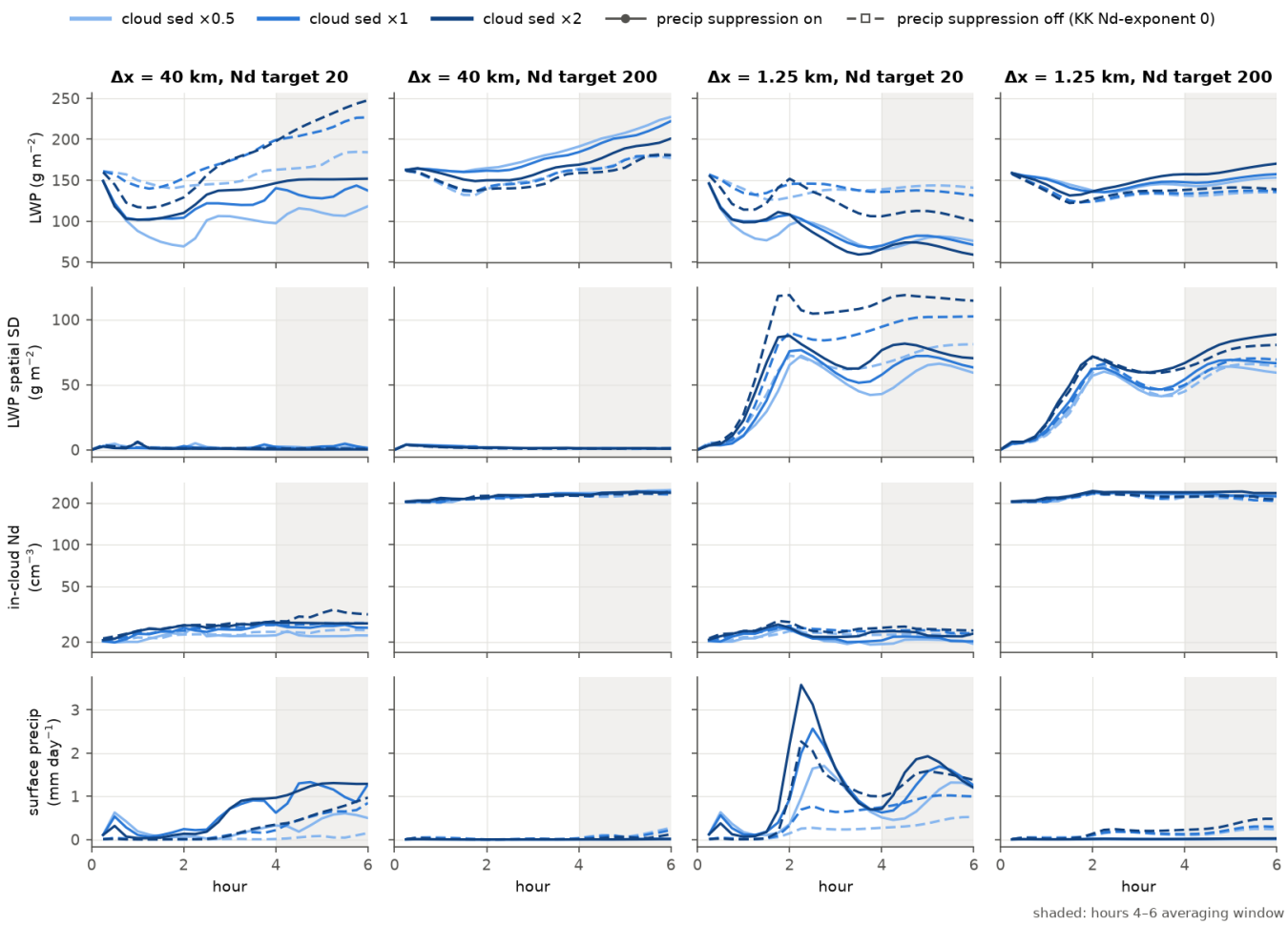


Fig. 3. 15-min time series at 40 km and 1.25 km, Nd 20 and 200. In-cloud Nd is the layer mean from the domain-mean profiles. The shaded band is the averaging window.

Data, code, paths (Perlmutter, world-readable)

- Runs: `/pscratch/sd/m/mahf708/dp_screamxx_sedps/sedps_<res>_nd<Nd>_sed<0p5|1|2>_ps<on|off>/run/`
- Analysis: `/pscratch/sd/m/mahf708/dp_screamxx_sedps/analysis/`
  - `summary_h4-6.csv`: one row per run.
  - `timeseries_15min.csv`: one row per run and 15-min step, run settings as columns, with LWP, LWP SD and three Nd measures.
  - `susceptibility.csv`, the figures, and `R/` (R outputs).
- Scripts: `/pscratch/sd/m/mahf708/scmlib/sed-precsupp-study/` (`analysis/` for Python, `R/` for R).
- R: `source("/pscratch/sd/m/mahf708/scmlib/sed-precsupp-study/R/sedps.R")`. Interpreter: `/pscratch/sd/m/mahf708/r_env/bin/Rscript`.
- Code: E3SM `mahf708/eamxx/p3-cloud-sed-factor` and scmlib `sed-precsupp-study`, both on github.com/mahf708.

Notes. Surface precipitation includes the cloud-droplet sedimentation flux. With suppression off, autoconversion matches the default rate at Nd 63. LWP SD is written online as a variance and its square root is taken afterwards. That is exact for the instantaneous 15-min snapshots used here. For time-averaged streams, the mean of the variance gives the RMS SD rather than the mean SD, so a native `sqrt` would help there.